

YUJIAO CHEN

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Google Scholar: scholar.google.com/citations?user=JlrC1jMAAAAJ | 2,880 citations | h-index: 18

RESEARCH AGENDA

AI governance researcher studying how interactions between AI systems create risks that cannot be identified by evaluating individual models alone. My research shows that deployment rules, incentives, and institutional design can fundamentally change collective AI behavior, suggesting that governments should evaluate AI systems, not just AI models. My work combines empirical AI research with experience deploying machine learning in safety-critical infrastructure (U.S. DOE), commercial systems (Google), and large-scale online marketplaces (Amazon), providing a practical perspective on what AI governance can realistically require.

RESEARCH & PROFESSIONAL EXPERIENCE

MIT — Postdoctoral Researcher

Cambridge, MA | Feb 2026 – Present

- **Research leadership** — leading an independent research program on AI governance and multi-agent safety.
- **Institutional design** — showed that AI agents that behave reasonably on their own can nevertheless produce harmful collective outcomes because of the incentives and rules governing their interactions. The work demonstrates that changing the institutional environment, not the models themselves, can restore cooperative behavior.
- **AI evaluation** — introduced “institutional red-teaming”, an approach for testing deployment rules rather than only AI models, suggesting a new direction for AI assurance and evaluation standards.
- **Trust between AI agents** — produced one of the first quantitative studies of how trust forms, breaks down, and recovers between AI agents, providing empirical evidence relevant to governing AI systems that increasingly cooperate with one another.
- **Decision-making under catastrophic risk** — showed that seemingly irrational AI behavior can emerge from rational optimization under catastrophic environments, highlighting how deployment conditions shape AI behavior.

Amazon — Applied Scientist in Machine Learning

New York City | Oct 2021 – Jan 2026

- **Reinforcement learning at scale** — built models that set pricing strategy and allocate products across Amazon’s advertising marketplace, where reinforcement learning systems made economically consequential decisions at scale.
- **Generative AI** — built LLM-based systems that automatically detect and explain anomalies in high-volume business metrics, catching and diagnosing problems earlier with far less manual investigation.
- **Experimentation** — designed and ran large-scale online experiments and causal-inference analyses, giving major business decisions a statistically rigorous evidence base.

Google — Data Scientist

New York City | Mar 2020 – Oct 2021

- **Machine learning control** — deployed machine learning systems that autonomously controlled commercial building energy systems in real-world operation.
- **ML infrastructure** — built the end-to-end real-time data pipelines that fed and monitored these deployed systems, with automated data-quality checks and alerting.

Pacific Northwest National Laboratory (U.S. DOE) — Postdoctoral Research Associate

Jul 2019 – Mar 2020

- **Reinforcement learning for critical infrastructure** — developed algorithms that help the power grid respond to emergencies (U.S. DOE ARPA-E program) — a setting where AI failure has physical, real-world consequences, resulting in two co-first-author papers in IEEE Transactions on Power Systems.
- **Applied analytics** — analyzed sensor data from federal buildings and smart-city infrastructure and delivered operational recommendations to the U.S. General Services Administration (GSA).

Harvard University — Doctoral Researcher & Research Associate

Cambridge, MA | 2014 – 2019

- **Reinforcement learning** — pioneered RL control of real-world energy systems, achieving up to 23% efficiency gains; the first-author paper became one of the most-cited works on RL for energy-system control.
- **Transfer learning** — developed methods that let models trained in one physical environment adapt quickly to new ones; published in peer-reviewed journal and conference venues.
- **Field research** — led multi-year measurement campaigns with hundreds of sensors and published the findings in top energy journals.

SELECTED RESEARCH CONTRIBUTIONS

16 journal articles, 4 conference proceedings, 4 recent works on AI safety & governance (2026) | 2,880 citations, h-index 18 | Full list on Google Scholar

- **Y. Chen**, “Individually Sensible, Collectively Harmful: Simulating Commons Failure in LLM Societies,” under review, 2026.
- **Y. Chen**, “Institutional Red-Teaming: Deployment Rules, Not Just Models, Causally Shape Multi-Agent AI Safety,” arXiv:2607.07695, 2026. [sole author]
- **Y. Chen**, “Trust Between AI Agents: Measuring Formation, Breakage, and Recovery, with Implications for Governing Multi-Agent Systems,” arXiv:2606.14923, 2026. [sole author]
- **Y. Chen**, “Prospect-Theory Behavior from Bellman Optimality in MDPs with Catastrophic States,” arXiv:2606.00970, 2026. [sole author]
- **Y. Chen**, L.K. Norford, H.W. Samuelson, A. Malkawi, “Optimal control of HVAC and window systems for natural ventilation through reinforcement learning,” Energy and Buildings, 2018. [370 citations]
- **Y. Chen**, Z. Tong, Y. Zheng, H. Samuelson, L. Norford, “Transfer learning with deep neural networks for model predictive control of HVAC and natural ventilation in smart buildings,” Journal of Cleaner Production, 2020. [278 citations]
- R. Huang*, **Y. Chen***, et al., “Learning and fast adaptation for grid emergency control via deep meta reinforcement learning,” IEEE Transactions on Power Systems, 2022. [66 citations]
- R. Huang*, **Y. Chen***, et al., “Accelerated derivative-free deep reinforcement learning for large-scale grid emergency voltage control,” IEEE Transactions on Power Systems, 2021. [67 citations]
- **Y. Chen**, Y. Zheng, H. Samuelson, “Fast adaptation of thermal dynamics model for predictive control using transfer learning with deep neural networks,” American Control Conference (ACC), 2020. [16 citations]
- **Y. Chen**, Z. Tong, A. Malkawi, “Investigating natural ventilation potentials across the globe: Regional and climatic variations,” Building and Environment, 2017. [281 citations]
- Z. Tong, **Y. Chen**, A. Malkawi, Z. Liu, R.B. Freeman, “Energy saving potential of natural ventilation in China: The impact of ambient air pollution,” Applied Energy, 2016. [337 citations]

EDUCATION

Harvard University	Cambridge, MA
<i>M.S. in Computational Science and Engineering</i>	2017 – 2018
<i>Doctor of Design (DDes) — dissertation on RL control of smart energy systems</i>	2016 – 2019
<i>Master in Design Studies (M.Des.), Energy and Environments</i>	2013 – 2014
University of Pennsylvania	Philadelphia, PA
<i>Master of Environmental Building Design (MSD-EBD)</i>	2012 – 2013
Zhejiang University	Hangzhou, China
<i>Bachelor of Architecture (B.Arch.)</i>	2007 – 2012

TEACHING

Harvard University — Teaching Fellow	2017 – 2019
• CS 109A/B (Introduction & Advanced Data Science): statistical ML, deep learning architectures (CNNs, RNNs, transformers, autoencoders), generative models, Bayesian methods, and NLP in Python. • AC 295 (Advanced Practical Data Science): scaling ML from prototype to production — containerization, Kubernetes, cloud deployment, transfer learning, and model distillation.	

AWARDS & SERVICE

- 1st Place, NYSERDA Tenant Energy Data Challenge (2021); Harvard CGBC Fellowship (2016–2019); Harvard Center for the Environment Fellowship (2018).
- Reviewer: American Control Conference (ACC), IEEE PES General Meeting, Applied Energy, Engineering Applications of Artificial Intelligence, and 8 other journals.